1.

Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

ANS:-

DNS Server and Zone Setup:

- Steps:
    - Install and configure BIND DNS Server on a Linux machine or VM.
    - Create a new DNS zone named musicmania.in.
    - Add the required zone configuration and save the settings.
    - Restart the DNS service to apply the changes.

Result: The DNS server was configured successfully, and a new DNS zone for musicmania.in was created.

2.

Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

ANS:-

Add and Verify A Record:

- A Record: www.musicmania.in

- Verification Details: 192.168.1.50 / nslookup www.musicmania.in

Result: The nslookup command returned 192.168.1.50, confirming that the A record was configured correctly.

3.

Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

ANS:-

CNAME Record Configuration:

- CNAME Record: play.musicmania.in
- Verification Details: www.musicmania.in / nslookup play.musicmania.in

Result: The lookup resolved play.musicmania.in to 192.168.1.50, confirming that the CNAME record works correctly.

4.

Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Hint: Include at least three different troubleshooting commands or checks in your documentation.

ANS:-

DNS Troubleshooting:

- **Troubleshooting Steps:**
    - Check the DNS configuration files to find any incorrect zone or record settings.
    - Check the DNS service status using: systemctl status bind9.
    - Restart the DNS service to apply the correct configuration: sudo systemctl restart bind9.
    - Test the domain resolution using: nslookup musicmania.in.
    - Verify the DNS response using: dig musicmania.in.

Result: These checks help identify configuration errors, restart the DNS service, and confirm whether musicmania.in is resolving correctly.